TEXORA · LECTURE SHEET

Frequency Distribution

Turning raw data into order — variables and classes, limits, boundaries and midpoints, the rules for building a table, and the diagrams that show a distribution at a glance: histogram, frequency polygon and ogive.

STAT-2109 · PROBABILITY AND STATISTICS · TOPIC 2 · LEVEL 2 TERM 1

1. Raw data and variables

Figures exactly as collected — 45 examination scripts with 45 separate marks written in the order they were marked — are **raw data**: correct, but unreadable as a whole. The characteristic being measured, here the mark, is a **variable**: it varies from student to student. Variables are of two kinds. A **discrete** variable takes separate, countable values — number of defects in a fabric piece (0, 1, 2, …), number of children in a family. A **continuous** variable can take any value within a range — mass, length, temperature, and marks treated on a scale. The **frequency** of a value or class is simply the number of observations falling in it. A frequency distribution is the complete statement of values (or classes) together with their frequencies.

2. Class intervals, limits, boundaries and midpoints

For many observations, values are grouped into **class intervals** such as 0–10, 10–20, 20–30. The two ends of a class are its **lower limit** and **upper limit**. Classes may be written in two ways:

- **Exclusive method:** 0–10, 10–20, 20–30 — an observation equal to 10 goes into the second class, not the first. The upper limit is excluded from its own class. This is the standard method for continuous data.
- **Inclusive method:** 0–9, 10–19, 20–29 — both limits are included in the class. A gap appears between classes, so before any calculation the classes must be converted to exclusive form.

Conversion of inclusive to exclusive classes. Find half the gap between the upper limit of one class and the lower limit of the next: for 0–9, 10–19 the gap is 10 − 9 = 1, half is 0.5. Subtract 0.5 from every lower limit and add 0.5 to every upper limit: 0–9 becomes **−0.5–9.5**, 10–19 becomes **9.5–19.5**. The adjusted ends are the **true class boundaries** (also called true limits). For exclusive classes the written limits already are the boundaries.

Two further quantities describe each class. The **class width** (or class interval, h) is upper boundary minus lower boundary. The **midpoint** (class mark), x, is the average of the two boundaries:

$$\text{Midpoint } x = (\text{lower boundary} + \text{upper boundary}) / 2$$

The midpoint is the working representative of its class: in grouped calculations (Topics 3 and 4) every observation in a class is treated as if it stood at the midpoint. This is an approximation, but a fair one when observations spread reasonably evenly within each class.

3. Rules for forming classes

- Classes must be exhaustive (cover every observation) and mutually exclusive (no observation in two classes).
- Equal class widths are preferred — unequal widths make diagrams and calculations misleading unless adjusted.
- Open-ended classes ("70 and above") should be avoided where possible; they have no midpoint and block most calculations.
- The number of classes is usually kept between about 5 and 20: too few hide the shape, too many leave ragged, near-empty classes.
- Class limits should be convenient numbers (multiples of 5 or 10) so the table is easy to read.

A widely used guide to the number of classes is **Sturges' rule**:

$$k = 1 + 3.322 \log_{10} N$$

WORKED EXAMPLE

For our data, $N = 45$: $\log_{10} 45 = 1.653$, so $k = 1 + 3.322 \times 1.653 = 1 + 5.49 =$ **$6.49 \approx 7$ classes**.

With marks running from 0 to 70, seven classes of width 10 — exactly the distribution used throughout Topics 2, 3 and 4.

4. The frequency table of the shared dataset

Marks out of 70 obtained by 45 students, grouped into the seven classes of width 10. Cumulative frequencies are given in both directions, with relative frequency and percentage.

CLASS	F	MIDPOINT X	CF (LESS THAN)	CF (MORE THAN)	RELATIVE FREQ. F/N	PERCENTAGE
0–10	2	5	2	45	0.044	4.4
10–20	6	15	8	43	0.133	13.3

20–30	10	25	18	37	0.222	22.2
30–40	14	35	32	27	0.311	31.1
40–50	8	45	40	13	0.178	17.8
50–60	4	55	44	5	0.089	8.9
60–70	1	65	45	1	0.022	2.2
Total	**45**	—	—	—	**1.000**	**100.0**

Reading the table: the **less-than cumulative frequency** at a class counts all observations below its upper boundary — 32 students scored less than 40. The **more-than cumulative frequency** counts all observations at or above its lower boundary — 27 students scored 30 or more. The **relative frequency**, f/N, expresses each class as a share of the whole; the shares sum to 1.000 (subject to rounding), and multiplying by 100 gives the percentage column. The shape is already visible: frequencies rise to a peak in 30–40 and fall away — a roughly bell-shaped distribution with a mild right-hand tail.

5. The histogram

A **histogram** represents a frequency distribution by rectangles standing on the class boundaries, with no gaps between them (the data are continuous) and with *area* proportional to frequency. With equal class widths, height may be read directly as frequency. The histogram — not the table — is what the eye judges: peak, spread, symmetry and outliers are seen at once.

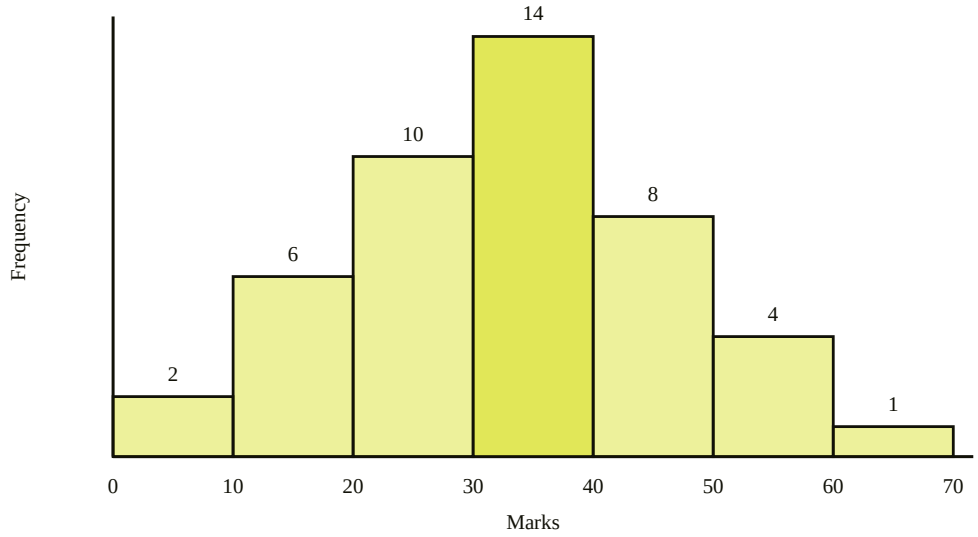


Figure 1 — Histogram of marks of 45 students (classes of width 10, frequencies 2, 6, 10, 14, 8, 4, 1)

6. The frequency polygon

A **frequency polygon** joins, by straight lines, the points plotted at each class midpoint with height equal to the class frequency. To close the figure to the axis, two extra points with zero frequency are added at the midpoints of the imaginary classes on either side — here at −5 and 75. The polygon and the histogram show the same shape; the polygon is handier when two distributions are compared on one diagram, since polygons can overlap where histograms cannot.

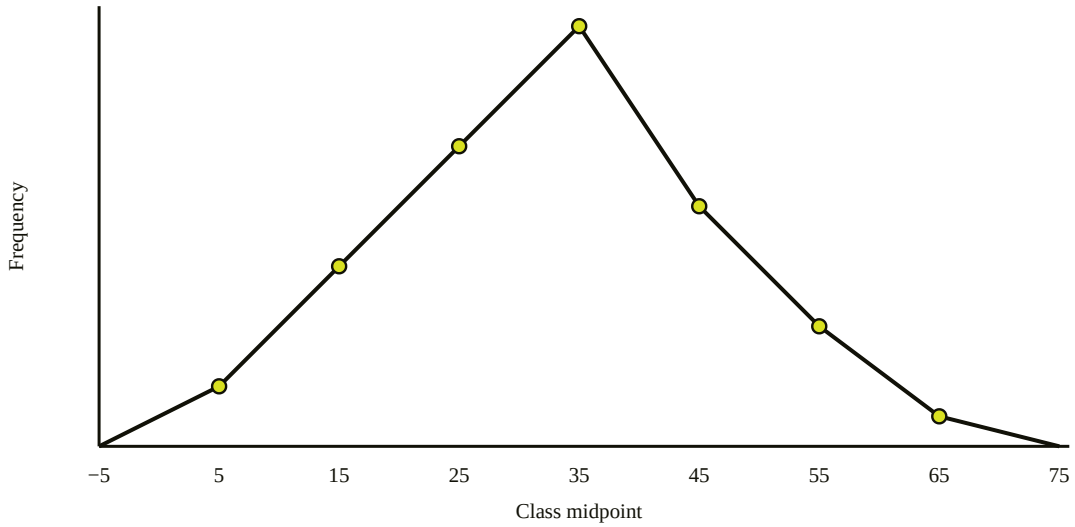


Figure 2 — Frequency polygon: midpoints 5 to 65 plotted against frequency, closed to the axis at −5 and 75

7. The ogive (cumulative frequency curve)

Plotting cumulative frequency against class boundaries gives an **ogive**. The **less-than ogive** plots less-than cumulative frequency against *upper* boundaries; for our data the points are (10, 2), (20, 8), (30, 18), (40, 32), (50, 40), (60, 44), (70, 45), starting from (0, 0). It climbs steadily and flattens at N = 45. The **more-than ogive** plots more-than cumulative frequency against *lower* boundaries and falls from 45 to 0. Ogives answer "how many below (or above) this value?" at a glance, and the value below which half the observations lie — the median — can be read off where the less-than ogive crosses cf = N/2 = 22.5 (see Topic 3).

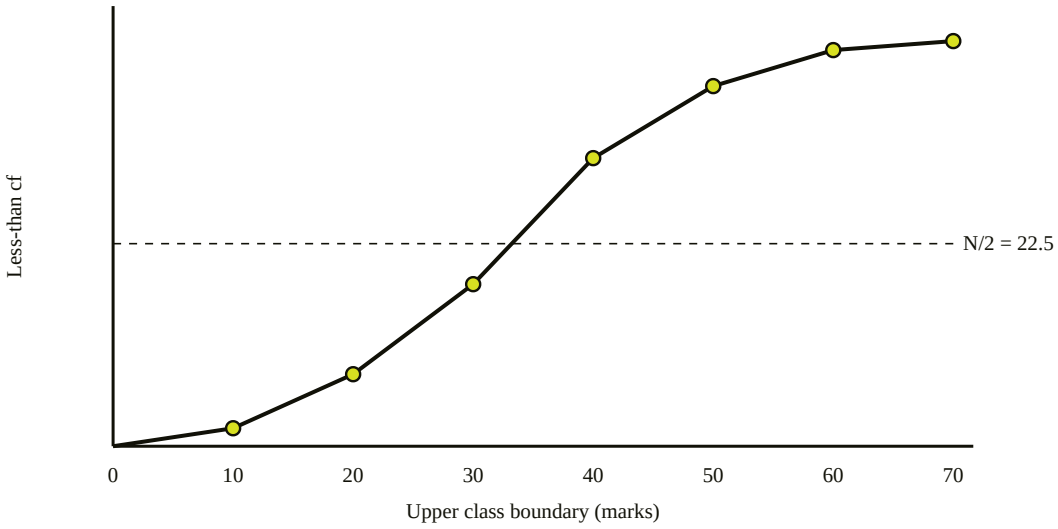


Figure 3 — Less-than ogive: cumulative frequencies 2, 8, 18, 32, 40, 44, 45 at upper boundaries 10 to 70

8. Other diagrams, and bar chart versus histogram

A **bar chart** serves categorical or discrete data: separate bars of equal width, with gaps between them, heights showing frequency or amount. A **pie chart** divides a circle in proportion to shares: each sector angle is (share) × 360°.

WORKED EXAMPLE

Pie chart angle for the modal class 30–40: its share is 14/45, so its sector angle is

$$14/45 \times 360° = 112°$$

The remaining sectors follow the same rule, and all seven angles sum to 360° — the pie chart's built-in check.

POINT	BAR CHART	HISTOGRAM
Data type	Categories or discrete values	Continuous, grouped data
Bars	Separated by gaps	Touching, no gaps
Meaning of a bar	Height shows the value	Area is proportional to frequency
Horizontal axis	Labels, in any order that suits	A true numerical scale, in order

Choosing the right diagram is part of honest presentation: a histogram for grouped continuous measurements, a bar chart for categories, a pie chart only when shares of one whole are the message.

Glossary

TERM	MEANING
Raw data	Observations in the form in which they were originally collected, unorganised.
Variable	A characteristic that takes different values for different observations; discrete or continuous.
Frequency	The number of observations falling in a value or class.
Class limits	The lower and upper ends written for a class interval.
Class boundaries	The true limits of a class after removing gaps between classes.
Midpoint (class mark)	The average of a class's two boundaries; the representative value of the class.
Cumulative frequency	The running total of frequencies, less-than or more-than a boundary.
Relative frequency	A class frequency divided by the total frequency, f/N.
Histogram	A diagram of touching rectangles over class boundaries, area proportional to frequency.
Ogive	A curve of cumulative frequency plotted against class boundaries.

Key points & summary

- Grouping turns raw data into a frequency distribution: classes with limits, true boundaries, width and midpoints.
- Inclusive classes (0–9, 10–19) are converted to exclusive form by adjusting each end by half the gap (0.5 here).
- Sturges' rule, $k = 1 + 3.322 \log_{10} N$, gives $k \approx 7$ classes for $N = 45$ — exactly the classes of the shared dataset.
- Less-than cf ends at N; more-than cf starts at N; relative frequencies sum to 1.000.
- The histogram shows shape through area; the frequency polygon joins midpoint points and closes at −5 and 75; the ogive reads cumulative counts and the median.
- Bar charts (gaps, categories) and histograms (no gaps, continuous scale) are not interchangeable.

Practice

Answer in your own words; brief model answers follow.

1. Convert the inclusive classes 20–29, 30–39, 40–49 into true class boundaries, and give the width and the midpoint of the first class.
2. Use Sturges' rule to find the number of classes for N = 200 observations.
3. From the table in Section 4, how many students scored (a) less than 50, (b) 40 or more? Show which column you used for each.
4. Why must a histogram have no gaps between its bars, while a bar chart must have them?
5. Find the pie chart angle for the class 20–30 of the shared dataset.
6. A less-than ogive is drawn for some data. Explain how you would read off approximately how many observations lie below 55.

Answer key

1. Gap = 30 − 29 = 1; half gap = 0.5. Boundaries: 19.5–29.5, 29.5–39.5, 39.5–49.5. Width = 10. Midpoint of first class = (19.5 + 29.5)/2 = 24.5.
2. $k = 1 + 3.322 \times \log_{10} 200 = 1 + 3.322 \times 2.301 = 1 + 7.64 = 8.64$, so about 9 classes.
3. (a) 40 students — the less-than cf at the class ending at 50. (b) 13 students — the more-than cf at the class beginning at 40.
4. The classes of continuous data meet at shared boundaries, so the rectangles must meet too; a gap would falsely suggest missing values. Bar chart bars stand for separate categories, and the gaps mark that separation.
5. Share = 10/45; angle = 10/45 × 360° = 80°.
6. Go up from 55 on the horizontal axis to the ogive, then across to the vertical (cumulative frequency) axis; the value read there is the approximate number of observations below 55.
